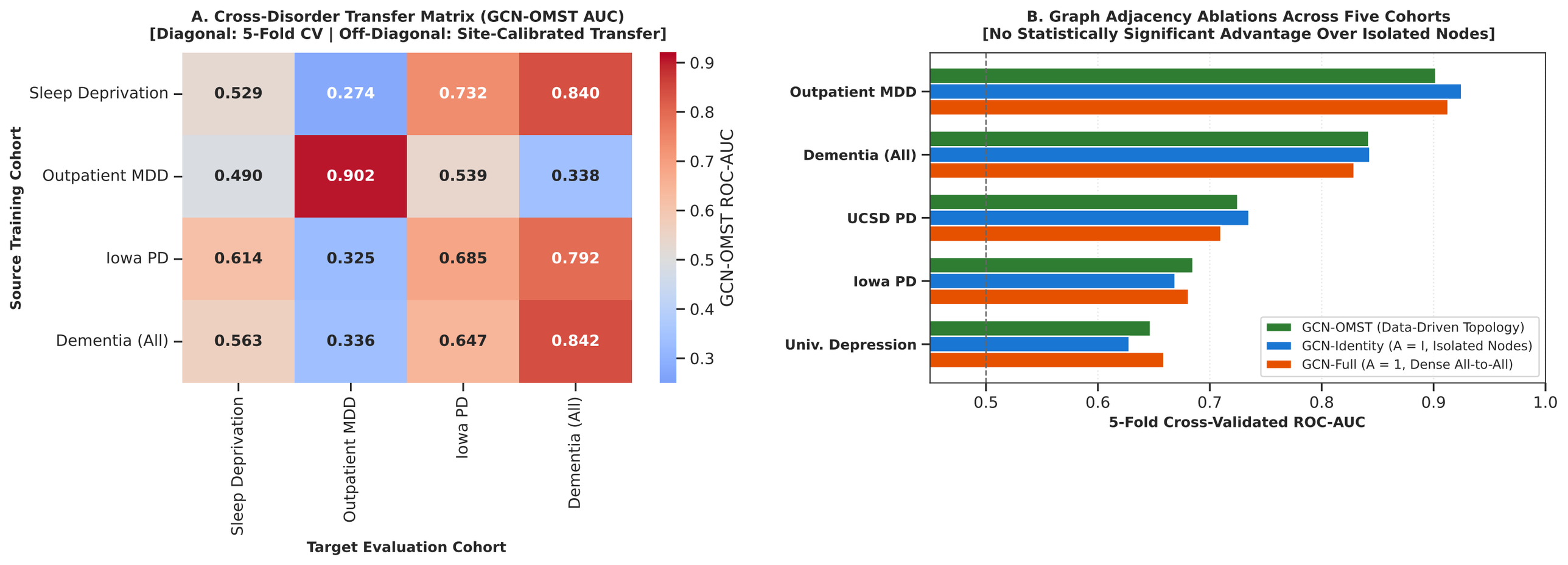
A. Cross-Disorder Transfer Matrix (GCN-OMST AUC)
[Diagonal: 5-Fold CV | Off-Diagonal: Site-Calibrated Transfer]
Source Training Cohort
Sleep Deprivation
Outpatient MDD
Iowa PD
Dementia (All)
0.529
0.274
0.732
0.840
0.490
0.902
0.539
0.338
0.614
0.325
0.685
0.792
0.563
0.336
0.647
0.842
Target Evaluation Cohort
GCN-OMST ROC-AUC
0.9
0.8
0.7
0.6
0.5
0.4
0.3
B. Graph Adjacency Ablations Across Five Cohorts
[No Statistically Significant Advantage Over Isolated Nodes]
Outpatient MDD
Dementia (All)
UCSD PD
Iowa PD
Univ. Depression
GCN-OMST (Data-Driven Topology)
GCN-Identity (A = I, Isolated Nodes)
GCN-Full (A = 1, Dense All-to-All)
0.5
0.6
0.7
0.8
0.9
1.0
5-Fold Cross-Validated ROC-AUC